Bao Nguyen	ECE332 Homework 4	05/25/26

1.) A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis? (8 points)

2.) The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \text{ for } 0 \le \theta \le \pi,$$

where θ is in radians. Determine: (15 points total)

a. the half power beamwidth, (5 points)

b. the pattern solid angle, (5 points)

c. the antenna directivity. (5 points)

3.) For a dipole antenna of length $l = \lambda/4$, (15 points total)
a. determine the directions of maximum radiation, (5 points)

b. obtain an expression for S_{max}, (5 points)

c. generate a plot of the normalized radiation pattern F (θ). (5 points)

4.) Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm. (5 points)

5.) A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km? (5 points)